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CSE C2 37

Lab 8: Form Processing using Django – Part II

Lab Assignments

urls.py (common):

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('cgpa/', include('cgpa.urls')),
    path('survey/', include('survey.urls')),
    path('register/', include('register.urls')),
    path('admin/', admin.site.urls),
]
```

- 1) Create a Register page and Success page with the following requirements:**
- i. Register page should contain four input TextBoxes for UserName, Password, Email id and Contact Number and also a button to submit. Make the username as compulsory field and other fields as optional.**
 - ii. On button click, Success page is displayed with message "Welcome {UserName}" and also his Email and Contact Number has to be displayed.**
 - iii. Use secure technique to send details to the Success page (Hint: use csrftoken) 4)**
- Design a website with two pages.**

forms.py:

```
from django import forms

class RegisterForm(forms.Form):
    username = forms.CharField(
        label="User Name",
        max_length=100,
        required=True
    )

    password = forms.CharField(
        label="Password",
        widget=forms.PasswordInput,
        required=False
    )
```

```
email = forms.EmailField(
    label="Email",
    required=False
)

contact = forms.CharField(
    label="Contact Number",
    max_length=10,
    required=False
)
```

views.py:

```
from django.shortcuts import render
```

```
from .forms import RegisterForm
```

```
# Create your views here.
```

```
def register(request):
```

```
    if request.method == "POST":
```

```
        form = RegisterForm(request.POST)
```

```
        if form.is_valid():
```

```
            username = form.cleaned_data['username']
```

```
            email = form.cleaned_data.get('email')
```

```
            contact = form.cleaned_data.get('contact')
```

```
            context = {
```

```
                'username': username,
```

```
                'email': email,
```

```
                'contact': contact
```

```
            }
```

```
            return render(request, "success.html", context)
```

```
    else:
```

```
        form = RegisterForm()
```

```
return render(request, "register.html", {'form': form})
```

urls.py:

```
from django.urls import path
```

```
from . import views
```

```
urlpatterns = [  
    path("", views.register, name='register'),  
]
```

register.html:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Register</title>
```

```
</head>
```

```
<body>
```

```
<form method="POST">
```

```
    {% csrf_token %}
```

```
<p>
```

```
    {{ form.username.label_tag }}<br>
```

```
    {{ form.username }}
```

```
</p>
```

```
<p>
```

```
    {{ form.password.label_tag }}<br>
```

```
    {{ form.password }}
```

```
</p>
```

```
<p>
  {{ form.email.label_tag }}<br>
  {{ form.email }}
</p>
```

```
<p>
  {{ form.contact.label_tag }}<br>
  {{ form.contact }}
</p>
```

```
<button type="submit">Submit</button>
```

```
</form>
```

```
</body>
```

```
</html>
```

success.html:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Success</title>
```

```
</head>
```

```
<body>
```

```
<h3>Welcome {{ username }}</h3>
```

```
{% if email %}
```

```
<p>Email: {{ email }}</p>
```

```
{% endif %}
```

{% if contact %}

<p>Contact Number: {{ contact }}</p>

{% endif %}

</body>

</html>

output:



A screenshot of a web browser window with a single tab titled 'Register'. The address bar shows the URL 'http://127.0.0.1:8000/register/'. The page contains a registration form with the following fields and values:

- User Name: Mathpal
- Password: (empty)
- Email: aditya@gmail.com
- Contact Number: 9999999999

At the bottom of the form is a 'Submit' button.



A screenshot of a web browser window with a single tab titled 'Success'. The address bar shows the URL 'http://127.0.0.1:8000/register/'. The page displays a confirmation message:

Welcome Mathpal
Email: aditya@gmail.com
Contact Number: 9999999999

2) "How is the book ASP.NET with c# by Vipul Prakashan?" Give the user three choice: i)Good ii)Satisfactory iii)Bad. Provide a VOTE button. After user votes, present the result in percentage using labels next to the choices

forms.py:

from django import forms

```
class SurveyForm(forms.Form):
```

```
    CHOICES = [
        ('good', 'Good'),
        ('satisfactory', 'Satisfactory'),
        ('bad', 'Bad'),
    ]
```

```
    choice = forms.ChoiceField(
        label="How is the book ASP.NET with C# by Vipul Prakashan?",
        choices=CHOICES,
        widget=forms.RadioSelect,
        required=True
    )
```

views.py:

from django.shortcuts import render, redirect

from .forms import SurveyForm

Create your views here.

```
good_votes = 0
satisfactory_votes = 0
bad_votes = 0

def survey(request):
    global good_votes, satisfactory_votes, bad_votes

    if request.method == "POST":
        form = SurveyForm(request.POST)

        if form.is_valid():
            choice = form.cleaned_data['choice']

            if choice == "good":
                good_votes += 1
            elif choice == "satisfactory":
                satisfactory_votes += 1
            elif choice == "bad":
                bad_votes += 1

            return redirect('result')
        else:
            form = SurveyForm()

    return render(request, "vote.html", {"form": form})

def result(request):
    total = good_votes + satisfactory_votes + bad_votes

    good_percent = (good_votes / total) * 100
    satisfactory_percent = (satisfactory_votes / total) * 100
```



```
bad_percent = (bad_votes / total) * 100
```

```
context = {  
    "good": good_percent,  
    "satisfactory": satisfactory_percent,  
    "bad": bad_percent,  
}
```

```
return render(request, "result.html", context)
```

urls.py:

```
from django.urls import path  
from . import views
```

```
urlpatterns = [  
    path("", views.survey, name='survey'),  
    path('result/', views.result, name='result'),  
]
```

vote.html:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Survey</title>
```

```
</head>
```

```
<body>
```

```
<form method="POST">
```

```
    {% csrf_token %}
```

```
    {{ form.as_p }}
```

```
<button type="submit">Vote</button>
</form>
```

```
</body>
</html>
```

result.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Survey Result</title>
</head>
<body>

<h3>How is the book ASP.NET with C# by Vipul Prakashan?</h3>

<p>Good: {{ good|floatformat:0 }}%</p>
<p>Satisfactory: {{ satisfactory|floatformat:0 }}%</p>
<p>Bad: {{ bad|floatformat:0 }}%</p>

<br>
<a href="{% url 'survey' %}">Return to Survey</a>

</body>
</html>
```

output:

Survey

x

+

←

→

↻

🔍

📄

🔗

http://127.0.0.1:8000/survey/

☆

🔍

📄

🔗

How is the book ASP.NET with C# by Vipul Prakashan?

☒ Good
☐ Satisfactory
☐ Bad

Survey Result

x

+

←

→

↻

🔍

📄

🔗

http://127.0.0.1:8000/survey/result/

☆

🔍

📄

🔗

How is the book ASP.NET with C# by Vipul Prakashan?

Good: 50%

Satisfactory: 50%

Bad: 0%

[Return to Survey](#)

3) Create a website with two pages. Page 1 has two TextBoxes (name and total marks) and one 'Calculate' Button as shown in the figure. On clicking the 'Calculate' Button, CGPA (total marks/50) along with the name should be displayed in the Page 2. Use Django sessions to store the information.

Name: Alex Total Marks: 450 Calculate Page 1	Welcome Alex Your CGPA is = 9 Page 2
---	---

forms.py:

```
from django import forms
```

```
class CGPAForm(forms.Form):  
    name = forms.CharField(  
        label="Name",  
        max_length=100,  
        required=True  
    )  
  
    total_marks = forms.FloatField(  
        label="Total Marks",  
        required=True  
    )
```

views.py:

```
from django.shortcuts import render, redirect  
from .forms import CGPAForm  
  
def page1(request):
```

```
if request.method == "POST":
    form = CGPAForm(request.POST)

    if form.is_valid():
        name = form.cleaned_data['name']
        total_marks = form.cleaned_data['total_marks']

        cgpa = total_marks / 50

        request.session['name'] = name
        request.session['cgpa'] = cgpa

        return redirect('page2')
    else:
        form = CGPAForm()

return render(request, "page1.html", {"form": form})
```

```
def page2(request):
    name = request.session.get('name')
    cgpa = request.session.get('cgpa')

    context = {
        "name": name,
        "cgpa": cgpa
    }

    return render(request, "page2.html", context)
```

urls.py:

```
from django.urls import path
from . import views
```

```
urlpatterns = [
    path('page1/', views.page1, name='page1'),
    path('page2/', views.page2, name='page2'),
]
```

page1.html:

```
<!DOCTYPE html>

<html>

<head>
    <title>Page 1</title>
</head>

<body>

<form method="POST">
    {% csrf_token %}

    <p>
        {{ form.name.label_tag }}
        {{ form.name }}
    </p>

    <p>
        {{ form.total_marks.label_tag }}
        {{ form.total_marks }}
    </p>

    <button type="submit">Calculate</button>
```

</form>

</body>

</html>

page2.html:

<!DOCTYPE html>

<html>

<head>

<title>Page 2</title>

</head>

<body>

{% if name and cgpa %}

<h2>Welcome {{ name }}</h2>

<h3>Your CGPA is = {{ cgpa|floatformat:2 }}</h3>

{% else %}

<p>No session data found.</p>

{% endif %}

Go Back

</body>

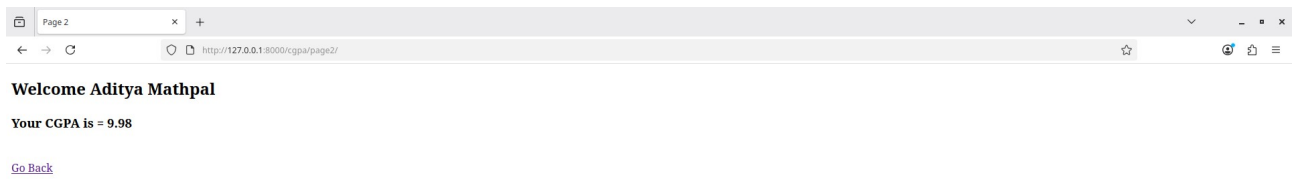
</html>

output:



A screenshot of a web browser window. The address bar shows the URL `http://127.0.0.1:8000/cgpa/page1/`. The page title is "Page 1". The form contains the following elements:

- A text input field labeled "Name:" with the value "Aditya Mathpal".
- A text input field labeled "Total Marks:" with the value "499".
- A button labeled "Calculate".



A screenshot of a web browser window. The address bar shows the URL `http://127.0.0.1:8000/cgpa/page2/`. The page title is "Page 2". The content of the page is:

Welcome Aditya Mathpal

Your CGPA is = 9.98

[Go Back](#)